Research Synopsis: Land-Use/Land-Cover Change in Pune District, 1990–2026

Title page

Proposed thesis title: Multi-Decadal Land-Use/Land-Cover Transformation in Pune District, Maharashtra (1990–2026): A Harmonised Multi-Sensor Earth-Observation Framework with Pre-Registered, Design-Based Validation

Field	Details
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Abstract

This research builds a carefully tested 30-year, 30 m record of land-use/land-cover (LULC) change for Pune District (about 15,650 km²). Free global land-cover products disagree by up to four times on Pune's built-up area, lack a reliable 1990 baseline and give no local error bars. The study combines Landsat 5/7/8/9, Sentinel-1 radar, a 30 m elevation model and CHIRPS rainfall on one fixed grid. Older Landsat data are corrected to the modern sensor with locally fitted pseudo-invariant-feature regression. Random Forest and temporal convolutional networks classify five land-cover classes, with calibrated probabilities and conformal prediction sets. Six pre-registered experiments (C1–C6) test training-label quality, sensor correction, era effects, regional transfer, calibration and the value of radar, against a blind, double-interpreted human sample of 641 points in 125 blocks. The district data foundation is complete (146 products, about 95 GB, every file fingerprinted); human labelling is the next stage. Expected outcomes are validated maps for 1990, 2000, 2010 and 2020, and area and change estimates with confidence intervals.

1. Introduction and background

Pune District has transformed since 1990. The city has spread along the Mutha–Mula corridor and highways into farmland and foothills; reservoirs such as Khadakwasla and Mulshi fluctuate with the monsoon; the eastern talukas face recurrent drought; and the Western Ghats escarpment is under development pressure. Planners, water managers and disaster agencies need to know how much land changed, where, and with what confidence.

Satellite Earth observation is the only practical way to reconstruct this consistently over decades: Landsat gives free 30 m data from the 1980s, and Sentinel-1 adds cloud-independent radar from 2014. The challenge is no longer producing a map but producing one that can be trusted, which requires consistent inputs across sensor generations and independent, statistically designed validation (Olofsson et al., 2014; Stehman and Foody, 2019).

2. Problem statement and research gap

No trustworthy, locally validated, multi-decadal 30 m record of land change exists for Pune District.

Gap	Evidence	Consequence
Global products disagree	Built-up area in 2020 differs up to 4×; three products agree on only 60 % of cells	Planners choose between incompatible numbers
No reliable 1990 baseline	Automatic labels map the 1990 city corridor as 50–74 % built-up vs 21 % in an independent product	Change since 1990 is overstated or unknown
Sensor discontinuity	Landsat-5 years read higher red and lower NDVI on stable targets than Landsat 8/9	Camera change mimics land change
Monsoon cloud	June–September optical composites cover under half the district in several years	Monsoon processes are invisible to optical sensors
Transfer gap	Macro-F1 ≈ 0.98 within a test area falls to 0.66–0.81 in unseen areas	Local accuracy does not hold district-wide

Most LULC studies also validate with random pixel splits, which inflate accuracy (Ploton et al., 2020), and do not fix their rules before seeing results. This research fills both the regional and the methodological gap.

3. Research question, hypotheses and objectives

Research question. To what extent can harmonised multi-sensor Earth-observation time series, evaluated against design-based human reference data, quantify multi-decadal land-cover change in Pune District, separate climatic fluctuation from lasting human conversion, and give defensible baselines for flood and environmental exposure?

No.	Hypothesis	Test
H1	Local PIF–RMA harmonisation removes sensor-transition artefacts well enough to detect real multi-decadal change.	C2, C3
H2	Temporal context, radar and terrain improve classification in complex peri-urban and semi-arid landscapes over single-date spectra.	C1, C6
H3	In unfamiliar or fast-changing regions, models are less reliable than they claim; spatial validation and conformal prediction expose this.	C4, C5

Objectives

1. Build a seasonally harmonised 30 m district record from Landsat 5–9, with Sentinel-1 radar for the recent era.
2. Collect an independent, stratified, double-interpreted human reference sample for 1990, 2000, 2010 and 2020.
3. Test, under pre-registered rules, which training labels, harmonisation and models give accurate, transferable maps.
4. Produce validated district maps and design-based area and change estimates with confidence intervals.
5. Analyse urban expansion, reservoir and water dynamics, agricultural stability and Western Ghats forest persistence.
6. Derive flood-exposure proxies from radar and terrain, as a foundation for future scenario modelling.

4. Study area

The study covers all of Pune District: 17,380,091 cells of 30 m (about 15,640 km²), on a 6,546 × 5,624-pixel grid in UTM zone 43N aligned to the native Landsat lattice. The district runs from the wet Western Ghats in the west, through the Pune metropolitan corridor, to the semi-arid Deccan plateau in the east; three CHIRPS rainfall regions (wet west, transition, dry east) structure the transfer tests. Exploratory work used three benchmark windows (5.8 % of the district): the Khadakwasla–Mutha urban corridor, the Baramati agricultural plain and the Mulshi–Western Ghats forest escarpment. The validation sample lies outside these windows plus 2 km.

5. Data

All data are open and accessed through the Microsoft Planetary Computer catalogue; every product's scene list is recorded.

Dataset	Resolution	Period	Role
Landsat 5 TM, 7 ETM+, 8 OLI, 9 OLI-2 (Collection 2 Level-2, 6 bands + QA)	30 m	18 years: 1990–92, 1998–2002, 2008–12, 2018–22	Historical backbone
Sentinel-1 RTC γ^0 (VV, VH)	~10 m → 30 m	2020 (107 scenes)	Cloud-independent structure and wetness
Copernicus GLO-30 DEM	30 m	Static	12 terrain layers (slope, HAND, TWI…)
CHIRPS rainfall	~5 km	1981 onward	Rainfall regions, drought context
WorldCover, Esri, GLC_FCS30D, GHSL, WSF, JRC water	10–30 m	Various	Automatic training labels and benchmarks

Reference tier	Size	Use
Tier A gold: blind human interpretation of very-high-resolution historical imagery (±2 years)	641 points × 4 epochs; 160 double-interpreted	Accuracy and area estimation only
T1: same protocol, training-eligible areas	699 points (525 train, 174 tuning); 140 double	Training option, model selection

Reference tier	Size	Use
Silver: unanimous WorldCover/Esri 2020–21, purity ≥ 78 %	8.16 million cells (46.9 %)	Training option only

6. Methodology

Every stage records its inputs, code version and SHA-256 fingerprints of its outputs, so products rebuild bit-identically.

1. **Data preparation:** pinned STAC scene lists; each scene read onto the fixed 30 m district grid (co-registration < 0.1 pixel); cloud, shadow and Landsat 7 gaps removed with QA_PIXEL, never interpolated; per-pixel median composites for dry (Jan–May), monsoon (Jun–Sep), post-monsoon (Oct–Dec) and annual periods, each with clear-observation count, valid mask and uncertainty layers.
2. **Harmonisation:** Landsat 7 mapped to Landsat 8 OLI by local RMA regression; a treatment family also maps Landsat 5 TM with the frozen transform c2_tm_pif_v1, fitted on stable water and forest targets in Pune. Its benefit is tested (C2), not assumed.
3. **Features:** per cell and year, 16 optical features (dry-season reflectance, NDVI, MNDWI, NDMI, NBR, texture, post-monsoon indices and seasonal change), 6 Sentinel-1 features and 12 terrain features, stored in two validated Zarr feature cubes.
4. **Reference sampling:** stratified sample in 125 independent 6 km blocks across 13 strata, ≥ 2 km from training data; two blind interpreters with adjudication. Gate G2: ≥ 80 % labelled (2020, 2000), ≥ 20 % double-interpreted, Cohen's κ ≥ 0.6.
5. **Classification and uncertainty:** Random Forest (300 trees) and TempCNN on five training-label options; temperature-scaled probabilities; conformal sets at 90 % nominal coverage; selection on the T1 tuning split only.
6. **Pre-registered evaluation:** protocol v2, frozen on 2 October 2026 before any label existed: five seeds, stratified block bootstrap (2,000 resamples), Benjamini–Hochberg correction (q = 0.05). A result counts only if its interval excludes zero, it survives correction and ≥ 4 of 5 seeds agree.
7. **Mapping and analysis:** validated maps for four epochs, design-based area and change estimates (Olofsson et al., 2014), and thematic analyses of urban growth, water, agriculture, forest and flood exposure.

Test	Question	Decisive evidence
C1	Does the training-label source change map accuracy?	64 paired tests on gold, four epochs, RF and TempCNN
C2	Does the local Landsat-5 correction improve 1990 and 2000?	Gold accuracy 1990/2000 and built-up share within independent products' range
C3	Does an era-aware model beat a single model in older eras?	Gold accuracy per era
C4	Does a model transfer to an unseen rainfall region?	Macro-F1 lower bound ⩾ 0.70 and gap ⩽ 0.10 in every region
C5	Are probabilities and conformal sets honest in new regions?	ECE ⩽ 0.05, coverage ⩾ 0.85, abstention helps
C6	Do radar and terrain add value to optical data?	Paired macro-F1 test, leave-one-region-out, 2020

C1 runs first and fixes the training option and model for C2–C6; the runner enforces this order.

7. Work completed so far

The computational foundation is complete; the confirmatory phase waits for human reference labels. Phase 1–3 results are exploratory (measured against automatic or product references).

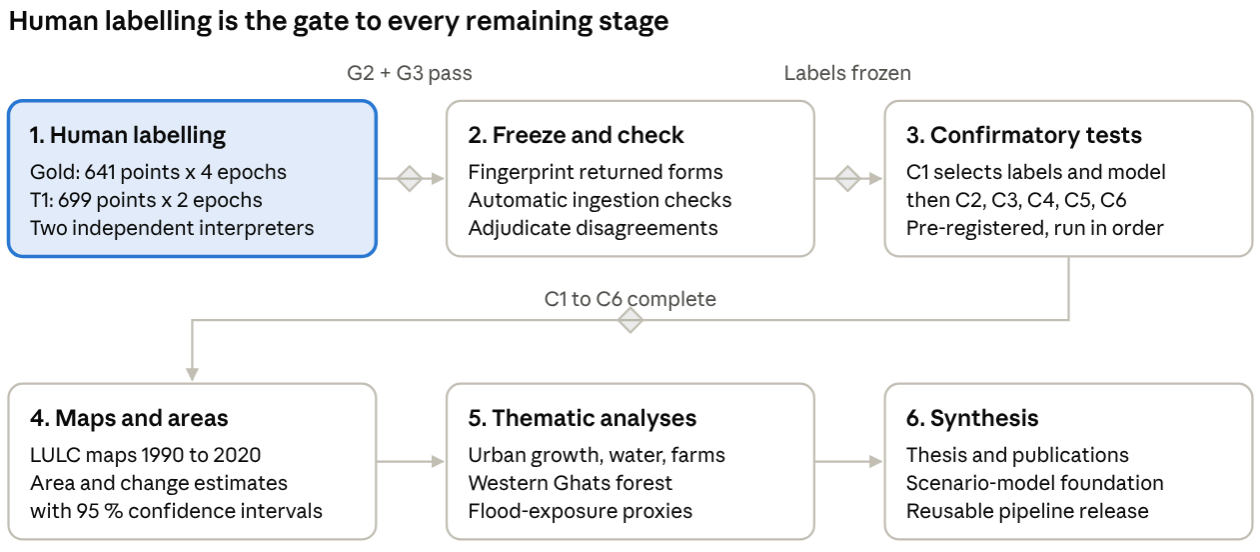
Phase	Main outcomes
1. Data foundation	37-year composites for three windows at 30 m and the district at 240 m; Sentinel-2 found 20–50 % brighter than Landsat in blue; monsoon optical unusable
2. Baseline models (35 experiments)	Macro-F1 ≈ 0.98 within a window, 0.66–0.81 in unseen windows; radar improved transfer (0.70–0.88); crop types not separable

Phase	Main outcomes
3. Systematic evaluation (63 experiments)	Several hypotheses rejected (sensor-transform transfer, domain adaptation, annual change timing); conformal coverage failed across landscapes; 0.979 silver agreement shown not to be accuracy
4. District design	Protocol v2 frozen; C1–C6 pre-registered; 641-point gold design; global products disagree up to 4×
5–6. Pilot, decisions, kits	30 m pilot epochs validated; local sensor correction and compute route decided; T1 design; blind interpreter kits
7. District build	72 Landsat + 44 corrected Landsat + 16 Sentinel-1 composites, 12 terrain layers, 2 validated feature cubes: 34,965 files (95.3 GB), all fingerprinted; 217 automated tests pass; C1–C6 code tested on synthetic data

Still pending: human gold 0 / 641, T1 0 / 699, confirmatory experiments 0 / 6.

8. Remaining work and indicative timeline

Every remaining stage depends on independent human reference labels. Labelling is therefore the critical path; all later stages are compute and analysis on data that already exists.



remaining work · 6 stages, 3 gates

Stage	Main tasks	Indicative duration (to confirm with supervisor)
1. Human labelling	Gold 2020 + 2000 first, then 2010 + 1990; T1 2020 + 2000; about 80–160 person-hours in total	2–3 months, depending on interpreter availability
2. Freeze and check	Freeze forms, automatic ingestion checks, adjudication, agreement (κ)	2–4 weeks
3. Confirmatory tests	Run C1, then C2–C6; family-wise multiple-testing correction	About 1 month
4. Maps and areas	District maps for four epochs; design-based area and change estimates	1–2 months
5. Thematic analyses	Urban growth modes, water–rainfall, agriculture, forest, flood exposure	3–4 months
6. Synthesis	Thesis writing, journal articles, data and code release	3–6 months

9. Expected outcomes

1. Answers to the six pre-registered questions (C1–C6), each supported, not supported or inconclusive, with confidence intervals.
2. Validated 30 m LULC maps of Pune District for 1990, 2000, 2010 and 2020, with per-pixel probabilities and prediction sets.
3. Design-based area and change estimates with 95 % confidence intervals, including a 1990→2020 transition matrix.
4. Thematic findings on urban growth, water, agriculture, Western Ghats forest and flood exposure.
5. A reusable human-validated reference dataset (641 points × 4 epochs) and an open, reproducible pipeline.
6. Three planned papers: the harmonised archive, the confirmatory evaluation, and Pune's land-change trajectories.

10. Significance and applications

The scientific novelty is a pre-registered, confirmatory test of choices that land-cover studies usually assume (label source, sensor correction, era effects, transfer, calibration, radar value), on a locally harmonised 30-year record graded by a blind, designed human sample. For Pune, it gives the first defensible, uncertainty-quantified land-change baseline since 1990.

Likely users include planning authorities (urban growth), water and irrigation departments (reservoir dynamics), disaster agencies and insurers (built-up land in flood-exposed terrain), agriculture and forest departments (cropland and forest persistence), ESG and due-diligence teams (historical baselines), and mapping companies (transfer and calibration audits). The framework uses only open data and runs on one laptop; the main cost of extending it is expert labelling time.

11. Limitations and research integrity

- **Limitations:** 30 m pixels mix covers; pre-2013 data allow change dating by epoch, not exact year; radar exists only from 2014; crop types are not separable; flood work gives exposure proxies, not flood depth; road–growth links are associations, not causes.
- **Risks:** interpreter availability (mitigated by labelling 2020 and 2000 first) and low inter-interpreter agreement (mitigated by protocol, adjudication and the G2 gate). Negative or inconclusive results remain publishable.
- **Integrity:** rules frozen before data; reference labels independent of model inputs; interpreters blind to products and models; automatic labels never used as reference; every file fingerprinted; failed experiments kept on record.

12. Proposed thesis structure

(1) Introduction · (2) Literature review · (3) Study area and data · (4) Building the harmonised archive · (5) Exploratory findings, Phases 1–3 · (6) Reference data and pre-registered design · (7) Confirmatory results, C1–C6 · (8) Pune's land change, 1990–2020 · (9) Discussion · (10) Conclusions and future work

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